Emphasis on Section 6, 7.

- Orthogonal and orthonormal sets and bases.
- Gram - Schmidt process, QR decomposition
- Orthogonal projection
- Least squares
- Spectral theorem for symmetric matrices
- Quadratic form
- Singular value decomposition

**Problem 1** Find orthogonal basis in

$H = \text{Span}\left\{ v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} \right\}$

$\dim H = 2$

$\{v_1, v_2\} \to \{u_1, u_2\}$

Gram - Schmidt:

$u_1 = v_1$

$u_1 \cdot u_1 = 3$

$v_2 \cdot u_1 = -1$

$$u_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4/3 \\ -2/3 \\ -2/3 \end{bmatrix}$$

$\frac{u_1}{\|u_1\|}, \frac{u_2}{\|u_2\|}$

$\|u_1\| = \sqrt{3}$

$$\|u_2\| = \sqrt{\frac{16}{9} + \frac{4}{9} + \frac{4}{9}} = \sqrt{\frac{24}{9}} = \sqrt{\frac{8}{3}} = \frac{2\sqrt{2}}{\sqrt{3}}$$

Orthonormal basis $\begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \begin{bmatrix} \sqrt{2}/\sqrt{3} \\ -\frac{\sqrt{2}}{2\sqrt{3}} \\ -\frac{\sqrt{2}}{2\sqrt{3}} \end{bmatrix}$

**Problem 2.** QR decomposition for $\begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}$

columns of Q form an orthonormal set

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{\sqrt{2}}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} & -\frac{\sqrt{2}}{2\sqrt{3}} \\ \frac{1}{\sqrt{3}} & -\frac{\sqrt{2}}{2\sqrt{3}} \end{bmatrix} \qquad R$$

upper triangular
positive entries
on the diagon

$$R = Q^T A = \begin{bmatrix} \sqrt{3} & -\frac{1}{\sqrt{3}} \\ 0 & \frac{2\sqrt{2}}{\sqrt{3}} \end{bmatrix}$$

**Problem 3.** Find matrix for orthogonal projection

$$\mathbb{R}^3 \longrightarrow \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} \right\}$$

If $H = \operatorname{Col} U$ Columns of $U$ are orthonormal

$U = Q$

$$UU^T = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{\sqrt{2}}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} & -\frac{\sqrt{2}}{23} \\ 1/\sqrt{3} & -\frac{\sqrt{2}}{2\sqrt{3}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{\sqrt{2}}{\sqrt{3}} & -\frac{\sqrt{2}}{2\sqrt{3}} & -\frac{\sqrt{3}}{2\sqrt{3}} \end{bmatrix} =$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Another orthonormal basis

$$U = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Least squares solution $A\underline{x} = \underline{b}$, best approximation.

$$A^T A \underline{x} = A^T \underline{b}$$ solve this system

**Problem 4.** Find the line $\boxed{y = \beta_0 + \beta_1 x}$ that best fits the data: points $(1,0), (2,1), (4,2), (5,3)$.

$(x, y)$

$\beta_0, \beta_1$ are unknown

$x = 1 \qquad \beta_0 + \beta_1 = 0$

$x = 2 \qquad \beta_0 + 2\beta_1 = 1$

$x = 4 \qquad \beta_0 + 4\beta_1 = 2$

$x = 5 \qquad \beta_0 + 5\beta_1 = 3$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

$25 + 16 + 4 + 1 = 46$

$$A^TA = \begin{bmatrix} 4 & 12 \\ 12 & 46 \end{bmatrix} \qquad A^T\underline{b} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 25 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 12 \\ 12 & 46 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 25 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 12 & 6 \\ 12 & 46 & 25 \end{bmatrix} \sim \begin{bmatrix} 4 & 12 & 6 \\ 0 & 10 & 7 \end{bmatrix} \sim \begin{bmatrix} 2 & 6 & 3 \\ 0 & 10 & 7 \end{bmatrix} \sim \begin{bmatrix} 10 & 0 & -6 \\ 0 & 10 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 30 & 15 \\ 0 & 10 & 7 \end{bmatrix}$$

$$\beta_1 = 0.7 \qquad \beta_0 = -0.6$$

$$y = -0.6 + 0.7x$$

**Theorem.** Any symmetric matrix $A$ is orthogonally diagonalizable, which means two equivalent conditions $A^T = A$

- $A = PDP^{-1}$ for some orthogonal matrix $P$ and diagonal matrix $D$

P has orthonormal column $P^TP = I$

- $A$ has an orthonormal eigenbasis.

**Theorem.** Quadratic form $Q(\underline{x}) = \underline{x}^T \underline{A}\, \underline{x}$, there is an orthogonal change of coordinates $\underline{y} = P\underline{x}$ such that $Q(\underline{y}) = \underline{y}^T D\, \underline{y}$. Principal axes theorem.

Problem 5. For the form $Q(\underline{x}) = 2x_1^2 + 6x_1x_2 - 6x_2^2$ find orthogonal change of coordinates such that $Q(y)$ has no cross term $y_1y_2$.

$$A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$$

$$\begin{vmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{vmatrix} = (\lambda-2)(\lambda+6) - 9 =$$

$$= \lambda^2 + 4\lambda - 12 - 9 =$$

$$= \lambda^2 + 4\lambda - 21 = (\lambda+7)(\lambda-3)$$

eigenvectors

$$\lambda_1 = -7$$

$$\lambda_2 = 3$$

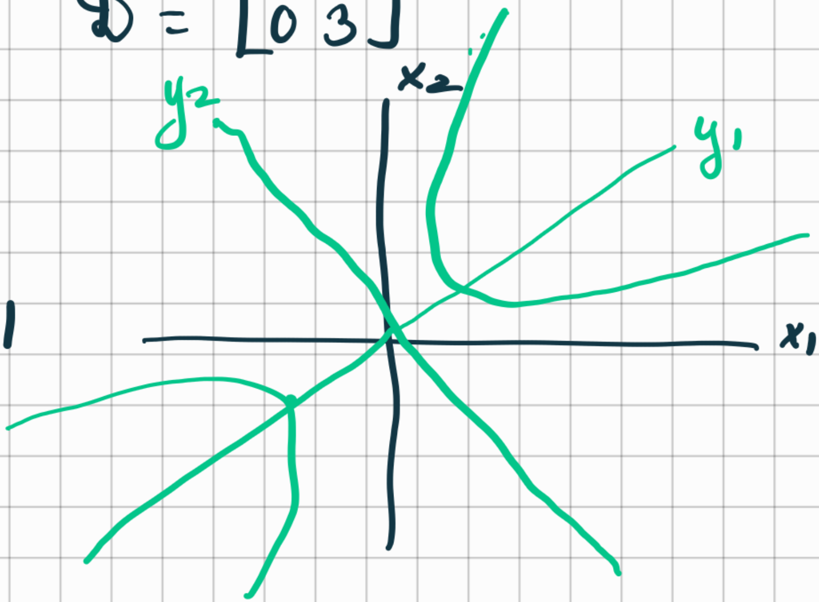
$\lambda = -7$ $\quad \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix} \underline{u}_1 = 0 \qquad \underline{u}_1 = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ -3 \end{bmatrix}$

$\lambda = 2$ $\quad \begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix} \underline{u}_2 = 0 \qquad \underline{u}_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$P = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & 3 \\ -3 & 1 \end{bmatrix} \qquad D = \begin{bmatrix} -7 & 0 \\ 0 & 3 \end{bmatrix}$

$Q(\underline{y}) = -7 y_1^2 + 3 y_2^2$

$Q(y)$ $\qquad Q(\underline{x}) = 1$




indefinite.

Types of form: Positive definite $\underline{x} \neq 0 \Rightarrow Q(\underline{x}) > 0$

semidefinite positive

$\lambda_1, \lambda_2, \lambda_3 \geq 0$

Negative definite $\underline{x} \neq 0 \Rightarrow Q(\underline{x}) < 0$

Indefinite : $Q(\underline{x}) > 0,\ Q(\underline{x}') < 0$

for some $\underline{x}, \underline{x}' \neq 0$.

Eigenvalues determine the type.

$$\lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$$

$\lambda_1, \lambda_2, \lambda_3$

eigenvalues of matrix

positive definite $\lambda_1, \lambda_2, \lambda_3 > 0$

negative definite $\lambda_1, \lambda_2, \lambda_3 < 0$

indefinite

Singular value decomposition

$$A = U \Sigma V^T$$

$A$ $m \times n$ matrix

$U$ orthogonal $m \times m$ matrix

$V$ orthogonal $n \times n$ matrix

$$\Sigma = \begin{array}{cc} & n \\ m & \begin{bmatrix} \sigma_1 & & & 0 \\ & \ddots & & \\ 0 & & \sigma_k & \\ 0 & & & 0 \end{bmatrix} \end{array}$$

singular values:

$A^T A$ $\quad \lambda_1 > \lambda_2 > \dots > \lambda_k > 0$

$$\sigma_i = \sqrt{\lambda_i}$$

# singular values

rk of matrix $A$

$$\operatorname{rk} A^T A = \operatorname{rk} A$$

Problem 6. Find SVD for $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

Step 1. $A^T A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

eigenvalues 0 and 1 singular values 1 $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

For matrix $V$ $[v_1 \ldots v_n]$ be orthonormal basis consisting of eigenvectors $A^T A$.

$U = \left[ \frac{A v_1}{\|A v_1\|} \quad \frac{A v_2}{\|A v_2\|} \quad \ldots \right]$

extend to any orthon

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{V^T}$$

Suppose $A$ be $n \times n$ matrix
$B$ is $n \times n$ matrix

suppose $A$ is invertible

If $AB$ is diagonalizable then $BA$ is diagonalizable.

$$\begin{matrix} 1 & 0 \\ 0 & 0 \end{matrix} \qquad \begin{matrix} 0 & 1 \\ 0 & 0 \end{matrix}$$

$$A^{-1}(AB)A = BA$$

$AB$
$BA$ similar

If $X, Y$ are similar and $X$ is diagonalizable then $Y$ is diagonalizable.